

Phase 4: Project Planning

1. Project Management Timeline

The development of the Human Development Index (HDI) Prediction System was divided into logical milestones or "Epics":

Epic 1: Data Collection & Preprocessing

Goal: Collect the Human Development Index dataset and prepare it for machine learning.

Tasks:

Obtain the HDI dataset.

Handle missing values and remove inconsistencies.

Perform data preprocessing and feature selection.

Conduct Exploratory Data Analysis (EDA) to understand relationships between variables.

Epic 2: Model Training & Evaluation

Goal: Train and evaluate Machine Learning models to predict the Human Development Index.

Tasks:

Split the dataset into training and testing sets.

Train different Machine Learning algorithms.

Evaluate model performance using metrics such as R^2 Score, Mean Absolute Error (MAE), and Mean Squared Error (MSE).

Select the best-performing model.

Outcome: The best-performing Machine Learning model was selected and saved as `hdi_model.pkl` for deployment.

Epic 3: Backend Development (Flask)

Goal: Develop the backend application to process user inputs and generate predictions.

Tasks:

Develop app.py using the Flask framework.

Load the trained model using Pickle.

Validate user inputs.

Integrate the Machine Learning model with the Flask application.

Create routes for prediction requests and result display.

Epic 4: Frontend Development

Goal: Design a simple and user-friendly web interface.

Tasks:

Develop HTML pages for the Home, Input, and Result screens.

Design responsive layouts using CSS.

Connect the input form with the Flask backend.

Display the predicted HDI score and development category.

Epic 5: Deployment & Documentation

Goal: Deploy the application online and prepare project documentation.

Tasks:

Create the requirements.txt file.

Organize the project folder structure.

Deploy the application on Render.

Test the deployed application.

Prepare project reports, documentation, and presentation materials.

2. Resource Allocation

Data Science:

Jupyter Notebook

Pandas

NumPy

Scikit-learn

Matplotlib

Machine Learning:

Python

Pickle (Model Serialization)

Web Development:

Flask

HTML5

CSS3

Bootstrap (Optional)

Development Environment:

Visual Studio Code (VS Code)

Version Control:

Git & GitHub

Deployment Platform:

Render

This follows the same structure as your sample while being customized for your Human Development Index (HDI) Prediction System.